Step 5. Check the solution. Substitute the solution into the original equation to make sure the result is a true statement.

8.4 Solve Equations with Fraction or Decimal Coefficients

- **Solve equations with fraction coefficients by clearing the fractions.**

  Step 1. Find the least common denominator of *all* the fractions in the equation.

  Step 2. Multiply both sides of the equation by that LCD. This clears the fractions.

  Step 3. Solve using the General Strategy for Solving Linear Equations.

Exercises

Review Exercises

Solve Equations using the Subtraction and Addition Properties of Equality

In the following exercises, determine whether the given number is a solution to the equation.

255. $x + 16 = 31,\ x = 15$

256. $w - 8 = 5,\ w = 3$

257. $-9n = 45,\ n = 54$

258. $4a = 72,\ a = 18$

In the following exercises, solve the equation using the Subtraction Property of Equality.

259. $x + 7 = 19$

260. $y + 2 = -6$

261. $a + \frac{1}{3} = \frac{5}{3}$

262. $n + 3.6 = 5.1$

In the following exercises, solve the equation using the Addition Property of Equality.

263. $u - 7 = 10$

264. $x - 9 = -4$

265. $c - \frac{3}{11} = \frac{9}{11}$

266. $p - 4.8 = 14$

In the following exercises, solve the equation.

267. $n - 12 = 32$

268. $y + 16 = -9$

269. $f + \frac{2}{3} = 4$

270. $d - 3.9 = 8.2$

271. $y + 8 - 15 = -3$

272. $7x + 10 - 6x + 3 = 5$

273. $6(n - 1) - 5n = -14$

274. $8(3p + 5) - 23(p - 1) = 35$

In the following exercises, translate each English sentence into an algebraic equation and then solve it.

275. The sum of -6 and m is 25.

276. Four less than n is 13.

In the following exercises, translate into an algebraic equation and solve.

277. Rochelle's daughter is 11 years old. Her son is 3 years younger. How old is her son?

278. Tan weighs 146 pounds. Minh weighs 15 pounds more than Tan. How much does Minh weigh?

279. Peter paid $\$9.75$ to go to the movies, which was $\$46.25$ less than he paid to go to a concert. How much did he pay for the concert?

280. Elissa earned $\$152.84$ this week, which was $\$21.65$ more than she earned last week. How much did she earn last week?

Solve Equations using the Division and Multiplication Properties of Equality

In the following exercises, solve each equation using the Division Property of Equality.

281. $8x = 72$

282. $13a = -65$

283. $0.25p = 5.25$

284. $-y = 4$

In the following exercises, solve each equation using the Multiplication Property of Equality.

285. $\frac{n}{6} = 18$ **286.** $\frac{y}{-10} = 30$ **287.** $36 = \frac{3}{4}x$

288. $\frac{5}{8}u = \frac{15}{16}$

In the following exercises, solve each equation.

289. $-18m = -72$ **290.** $\frac{c}{9} = 36$ **291.** $0.45x = 6.75$

292. $\frac{11}{12} = \frac{2}{3}y$ **293.** $5r - 3r + 9r = 35 - 2$ **294.** $24x + 8x - 11x = -7 - 14$

Solve Equations with Variables and Constants on Both Sides

In the following exercises, solve the equations with constants on both sides.

295. $8p + 7 = 47$ **296.** $10w - 5 = 65$ **297.** $3x + 19 = -47$

298. $32 = -4 - 9n$

In the following exercises, solve the equations with variables on both sides.

299. $7y = 6y - 13$ **300.** $5a + 21 = 2a$ **301.** $k = -6k - 35$

302. $4x - \frac{3}{8} = 3x$

In the following exercises, solve the equations with constants and variables on both sides.

303. $12x - 9 = 3x + 45$ **304.** $5n - 20 = -7n - 80$ **305.** $4u + 16 = -19 - u$

306. $\frac{5}{8}c - 4 = \frac{3}{8}c + 4$

In the following exercises, solve each linear equation using the general strategy.

307. $6(x + 6) = 24$ **308.** $9(2p - 5) = 72$ **309.** $-(s + 4) = 18$

310. $8 + 3(n - 9) = 17$ **311.** $23 - 3(y - 7) = 8$ **312.** $\frac{1}{3}(6m + 21) = m - 7$

313. $8(r - 2) = 6(r + 10)$ **314.** $5 + 7(2 - 5x) = 2(9x + 1) - (13x - 57)$ **315.** $4\,(3.5y + 0.25) = 365$

316. $0.25(q - 8) = 0.1\,(q + 7)$

Solve Equations with Fraction or Decimal Coefficients

In the following exercises, solve each equation by clearing the fractions.

317. $\frac{2}{5}n - \frac{1}{10} = \frac{7}{10}$ **318.** $\frac{1}{3}x + \frac{1}{5}x = 8$ **319.** $\frac{3}{4}a - \frac{1}{3} = \frac{1}{2}a + \frac{5}{6}$

320. $\frac{1}{2}(k + 3) = \frac{1}{3}(k + 16)$

In the following exercises, solve each equation by clearing the decimals.

321. $0.8x - 0.3 = 0.7x + 0.2$ **322.** $0.36u + 2.55 = 0.41u + 6.8$ **323.** $0.6p - 1.9 = 0.78p + 1.7$

324. $0.10d + 0.05(d - 4) = 2.05$

Practice Test

325. Determine whether each number is a solution to the equation.
$3x + 5 = 23$.
ⓐ 6 ⓑ $\frac{23}{5}$

In the following exercises, solve each equation.

326. $n - 18 = 31$

327. $9c = 144$

328. $4y - 8 = 16$

329. $-8x - 15 + 9x - 1 = -21$

330. $-15a = 120$

331. $\frac{2}{3}x = 6$

332. $x + 3.8 = 8.2$

333. $10y = -5y + 60$

334. $8n + 2 = 6n + 12$

335. $9m - 2 - 4m + m = 42 - 8$

336. $-5(2x + 1) = 45$

337. $-(d + 9) = 23$

338. $\frac{1}{3}(6m + 21) = m - 7$

339. $2(6x + 5) - 8 = -22$

340. $8(3a + 5) - 7(4a - 3) = 20 - 3a$

341. $\frac{1}{4}p + \frac{1}{3} = \frac{1}{2}$

342. $0.1d + 0.25(d + 8) = 4.1$

343. Translate and solve: The difference of twice x and 4 is 16.

344. Samuel paid $\$25.82$ for gas this week, which was $\$3.47$ less than he paid last week. How much did he pay last week?